

Shivam Sinha

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EDUCATION

Northeastern University, Boston, MA

Master of Science in Informatics - *Advanced cloud computing, AWS Cloud Architecture, Python and Analytics, Network protection and security, Cloud analytics*

Sep 2023 - Mar 2025

GPA 3.9

Vellore Institute of Technology, Vellore, India

B.Tech in Electronical Engineering – *Advanced Java Programming, Control Systems, DBMS, OOPS, Data Structures and Algorithms, Robotics and Control, Neural Networks*

Jul 2017 - Jul 2021

WORK EXPERIENCE

Software Engineer –Site Reliability | Shell PLC - Wipro Limited, Belgium, Europe (Remote)

Aug 2021 - Jul 2023

- Designed & managed **Jenkins CI/CD** pipelines integrated with **GitLab & Kubernetes**, automating deployment of microservices across Dev & Production clusters, improving release frequency by 25% & reducing deployment failures by 30%
- Built & containerized **Python/Node.js** automation scripts using **Docker** & deployed them on **Kubernetes**, integrating **REST APIs** to eliminate 200+ hours of manual effort weekly for a 25-member audit team & increasing operational efficiency by **20%**
- Engineered a real-time AWS cost optimization architecture using **AWS Config, EventBridge, Lambda, & Terraform (IaC)** to automate detection & remediation of idle resources & cost anomalies, achieving 20% monthly AWS cost reduction
- Created **end-to-end observability** for **CI/CD pipelines & microservices** using **Prometheus, Grafana, and CloudWatch**, enabling real-time visibility into performance, failure trends, & resource utilization—resulting in 15% faster issue resolution
- Collaborated with **cross-functional teams** to design & deploy **highly available** systems running **Postgres, & MongoDB**

Platform & Automation Engineer | Wipro Limited, Bengaluru, India

Jul 2021 - Aug 2021

- Managed server deployments, executed **script automation** and **streamlined updates on VMs**, reducing downtime by 20%
- Implemented **GitOps** workflows with **ELK & log analysis**, resolving deployment issues 23% faster via **automated rollbacks**
- Configured & maintained **Linux** environments (**Ubuntu, RHEL, CentOS**) with **GitLab, ServiceNow, & SDLC** tools, ensuring 99.9% uptime, security patch compliance, and streamlined automated deployment workflows using **Ansible**

SKILLS

Programming & Scripting Languages: Python, Java, LINUX, C++, Node.js, YAML

Cloud Platforms & Services: AWS (EC2, S3, Lambda, IAM, CloudWatch, DynamoDB, Aurora, SageMaker), Azure, GCP

DevOps & IaC: Terraform, CloudFormation, Ansible, Docker, Kubernetes, Jenkins, GitLab CI/CD, Git, SVN

Monitoring & Analytics Tools: Amazon CloudWatch, Datadog, Prometheus, Grafana, ELK Stack, Power BI, New Relic

Frameworks & Libraries: .NET, Selenium, NumPy, Pandas, TensorFlow, Keras, LangChain, MongoDB, Flask, Express.js

Databases & Data Technologies: SQL, PostgreSQL, AWS Aurora, RDBMS, MongoDB, Redis, Apache Kafka

Technologies & APIs: RESTful APIs, AWS SDK/CLI, GraphQL, IoT, Natural Language Processing (NLP)

Additional Skills: Infrastructure Implementation, Cloud Security Best Practices, Scrum Methodologies, Performance Optimization

CERTIFICATIONS

AWS Certified Cloud Practitioner – Amazon Certification

Dec 2024

AWS Certified Solutions Architect - Associate – Amazon Certification

May 2025

PROJECTS

AutoRes-Automated Disaster Recovery Infrastructure | EC2, Kubernetes, Jenkins, Prometheus, Grafana, Terraform

Apr 2025

- Built a self-healing, disaster-resilient infrastructure on **AWS** using **EC2, Kubernetes, Jenkins, & Lambda**; **automated rollback** & recovery from updates, network failures, & deletions, improving uptime by 99% & reducing manual effort by 90%

Serverless Machine Learning | Amazon Rekognition, AWS Lambda, AWS Step Functions, DynamoDB, EventBridge, SageMaker

Feb 2025

- Built a serverless image recognition system using **Amazon Rekognition** (98% accuracy), automated with **AWS Step Functions & EventBridge**, using **AWS Lambda** to process images from **S3** & store results in **DynamoDB** with 99.9% uptime
- Prof. Carmen Taglienti | Northeastern University

AI-Powered Talent Matching System | NLP, Flask, Scikit-learn, Pandas, AWS, JavaScript, Postman, LangChain

Aug 2024

- Developed a 92% accurate **MLP model** to match job roles to employee skills using **Flask APIs, Pandas, and Scikit-learn**; deployed on **AWS EC2**, tested via **Postman**, integrated with full-stack UI, enabling 50% faster talent screening

Prof. Zhi (Richard) He | Northeastern University